آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Teaching and Learning Resources for Grade IX Physics

Recommended Key Textbook:

Physics 9, Punjab Curriculum and Textbook Board (2023), Lahore

Additional Recommended Resource Material				
Topic No.	**Topic Title**	**Reference Book**	**Topic in the Reference Book**	**Website/ Link**
1	Measurements	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 4: Measuring	https://asana.com/resources/accuracy-vs-precision https://manoa.hawaii.edu/exploringourfluidearth/physical/world-ocean/map-distortion/practices-science-precision-vs-accuracy https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 3: Matter and Molecules	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 1: Measurements	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 1: Measurement of Physical Quantities	
		Caravan Book House (2021), *Physics 9*	Unit 1: Physical Quantities and Measurements.	

FOR ANNUAL EXAMINATION 2026 AND ONWARDS

		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 1: Measurements	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit: 2 Foundations of Measurement in Physics	

FOR ANNUAL EXAMINATION 2026 AND ONWARDS

Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
2	Kinematics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 8: Motion	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 5: Motion and Energy	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 3: Speed, Velocity and Acceleration Unit 5: Vectors	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 2: Speed Velocity and Acceleration	
		Caravan Book House (2021), *Physics 9*	Unit 2: Kinematics.	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 2: Kinematics	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 4: Kinematics	
3	Dynamics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 5: Forces	http://phet.colorado.edu/ http://www.123physics.com
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE*	Unit 5: Motion and Energy	

FOR ANNUAL EXAMINATIONS 2025 AND ONWARDS

		Physics, (3'd Ed) (2014). Hodder Edition.		http://www.physicsclassroom.com/class
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 2: Force of Gravity, Weight and Friction Unit 4: Newton's Laws of Motion	https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 3: Force	
		Caravan Book House 2021, *Physics 9*	Unit 3: Dynamics	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 3: Dynamics	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 5: Dynamics	

FOR ANNUAL EXAMINATION 2025 AND ONWARDS

Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
4	Turning Effect of Forces	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 5: Forces	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 6: Turning Forces	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 5: Turning Effect of Forces	
		Caravan Book House (2021), *Physics 9*	Unit 4: Turning Effect of Forces	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 4: Turning Effect of Forces	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 6: Torque and Stability	
5	Deformation of Solids	Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 3: Matter and Molecules	http://phet.colorado.edu/ http://www.123physics.com

FOR ANNUAL EXAMINATION 2025 AND ONWARDS

		Caravan Book House 2021, *Physics 9*	Unit 7: Properties of matter	http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 5: Force and matter	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 8: Deformation of Solids	
6	Work and Energy	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 7: Machines	http://phet.colorado.edu/ http://www.123physics.com http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 7: Work, Energy and Power	
		Caravan Book House 2021, *Physics 9*	Unit 6: Work and energy	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 8: Energy Sources and Transfer of Energy	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 7: Work, Energy and Power	

FOR ANNUAL EXAMINATION 2026 AND ONWARDS

Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
7	Pressure	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 6: Pressure	http://phet.colorado.edu/ https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 10: Pressure in Liquids and Gases	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 7: Pressure	
		Caravan Book House 2021, *Physics 9*	Unit 7: Properties of Matter	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 7: Properties of Matter	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 9: Pressure	
8	Thermal Physics	John Avison, (2'd Ed) (2014). *The World of Physics*. Nelson.	Unit 10: Heat and Matter	http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 6: Heat and Energy	
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 15: Expansion of Solids and Liquids	

FOR ANNUAL EXAMINATIONS 2026 AND ONWARDS

		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Section B: Thermal Physics	
		Caravan Book House 2021, *Physics 9*	Unit of 8: Thermal properties of matter	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 09: Thermal properties of Matter	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 10: The Science of Temperature	

FOR ANNUAL EXAMINATION 2026 AND ONWARDS

Topic No.	Topic Title	Reference Book	Topic in the Reference Book	Website/ Link
9	Transfer of Thermal Energy	Duncan, Tom and Kennett, Heather. South Asia Edition, *Cambridge IGCSE Physics*, (3'd Ed) (2014). Hodder Edition.	Unit 6: Heat and Energy	http://www.physicsclassroom.com/class https://ebooks.stbb.edu.pk/english/class-ix?sfnsn=scwspwa
		Abbot, A. F. (1989). *Physics*. India: Heinemann Educational Publishers.	Unit 17: Transmission of Heat	
		Charles Chew, Leong See Cheng, Chow Siew Foong. *Physics A Course of O Level*, Second Edition (2004).	Unit 12: Transfer of thermal energy	
		Caravan Book House 2021, *Physics 9*	Unit 9: Transfer of Heat	
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 9: Thermal Properties of Matter	
10	Nature of Science	Caravan Book House 2021, *Physics 9*	Unit 1: Physical Quantities and Measurement	https://www.britannica.com/science/scientific-method
		Sindh Textbook Board (2021). *The Textbook of Physics for Classes IX*. Jamshoro: Sindh Textbook Board.	Unit 1: Physical Quantities and Measurement	
		CANTAB Publisher, Lahore. *Model Textbook of Physics for Grade 9.* Based on National Curriculum of Pakistan 2022-23	Unit 1: Nature of Science	

FOR ANNUAL EXAMINATION 2026 AND ONWARD

Note: This resource list has been prepared primarily for teachers. While it can be shared with students, students should not be required to buy multiple books. Schools are encourages to stock these books in the library. Moreover, these are only suggestions which have been compiled for the ease of teachers and students; schools are encouraged to use other resources for teaching and learning as well, as long as they are in line with the student learning outcomes (SLOs) mentioned in AKU-EB syllabi. Unless specified, AKU-EB does not endorse any of these books or websites. You are advised to use an ad-blocker while accessing the websites. In case any website is not functional for any reason, you may inform us at examination.board@aku.edu for an alternative or search material via any search engine. If you have any query, please contact us via email.

FOR ANNUAL EXAMINATION 2026 AND ONWARDS